

## 2-Mark Questions & Answers -- Data Structures Module 3: Linked Lists

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### Q1. What is a Linked List State its node structure.

A Linked List is a dynamic linear data structure composed of nodes allocated in non-contiguous memory, where each node contains Data and a Pointer (next) storing the address of the subsequent node.

### Q2. Differentiate between Array and Linked List.

| Feature                  | Array                                    | Linked List                             |
|--------------------------|------------------------------------------|-----------------------------------------|
| **Memory Allocation**    | Static / Contiguous memory.              | Dynamic / Non-contiguous memory on heap |
| **Insertion / Deletion** | Slow $O(N)$ (requires shifting elements) | **Fast $O(1)$ (updating pointer links)  |
| **Element Access**       | Direct random access $O(1)$ .            | Sequential access $O(N)$ .              |

### Q3. What is a Doubly Linked List (DLL)

A Doubly Linked List has nodes containing two pointers: prev (pointing to predecessor) and next (pointing to successor), allowing bidirectional traversal.

### Q4. What is a Circular Linked List

A Circular Linked List is a linked list where the next pointer of the last node points back to the Head node instead of NULL.

### Q5. How is a Stack implemented using a Singly Linked List

Pushing and popping elements are performed at the Head of the Linked List in  $O(1)$  time (push: insert at beginning, pop: delete from beginning).